

Atlas Vitae University Hospital

Enterprise Network Infrastructure Technical Implementation Report



Atlas Vitae

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1 Executive Summary

The **Atlas Vitae University Hospital Enterprise Network Infrastructure** was designed and implemented to provide a secure, scalable and highly available network capable of supporting the operational requirements of a modern university hospital.

The primary objective of this implementation was to build an enterprise-grade network infrastructure based on Cisco technologies while following industry best practices for hierarchical network design, redundancy, traffic segmentation, service availability and network security.

The implemented infrastructure integrates multiple network services into a single cohesive environment, allowing administrative departments, medical staff, diagnostic systems, voice communications, wireless users and enterprise services to operate efficiently while maintaining strict logical separation between network segments.

The network architecture follows the traditional Cisco three-layer hierarchical model, consisting of Core, Distribution and Access layers. Dynamic routing is performed using the Open Shortest Path First (OSPF) protocol, while gateway redundancy is achieved through the Hot Standby Router Protocol (HSRP). Link redundancy and increased bandwidth between critical network devices are provided through EtherChannel using the Link Aggregation Control Protocol (LACP).

Several enterprise services were implemented as part of the infrastructure, including Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), Network Address Translation (NAT), Port Address Translation (PAT), File Transfer Protocol (FTP), electronic mail services, centralized logging (Syslog), Network Time Protocol (NTP), Cisco Call Manager Express (CME), Voice over IP (VoIP), Quality of Service (QoS) and Secure Shell (SSH) for secure remote administration.

Network security was reinforced through extensive Access Control Lists (ACLs), ensuring controlled communication between departments, protecting management resources and restricting access to critical internal services while maintaining public accessibility to external web services.

The entire infrastructure was validated through functional testing, confirming correct operation of routing protocols, redundancy mechanisms, network services, security policies, Internet connectivity, voice communications and service availability under both normal and simulated failure conditions.

This document provides a complete technical overview of the implemented infrastructure, describing its architecture, configuration principles, deployed services, security mechanisms, validation procedures and recommendations for future expansion.

2 Project Overview

2.1 Purpose

The purpose of this project was to design and implement a reliable, secure and scalable enterprise network infrastructure capable of supporting the operational requirements of Atlas Vitae University Hospital.

The infrastructure was developed following Cisco Enterprise Campus Design principles, ensuring high availability, efficient traffic segmentation and centralized network management.

2.2 Objectives

The implementation was designed to achieve the following objectives:

- Provide a hierarchical enterprise network architecture.
- Ensure network redundancy and high availability.
- Segment hospital departments using VLANs.
- Deploy essential enterprise network services.
- Secure internal resources through Access Control Lists (ACLs).
- Support Voice over IP (VoIP) communications.
- Provide wired and wireless connectivity.
- Allow secure Internet access.
- Simplify network administration and future expansion.

2.3 Scope

The implemented infrastructure includes:

- Core, Distribution and Access layers.
- Enterprise routing using OSPF.
- Gateway redundancy using HSRP.
- EtherChannel (LACP).
- DHCP and DNS services.
- NAT and PAT.
- Cisco Call Manager Express (CME).
- VoIP.
- QoS.
- SSH remote management.

- ACL-based security.
- Public and internal web services.
- Centralized monitoring and logging.

3 Network Architecture

3.1 Overview

The Atlas Vitae University Hospital network was designed following the Cisco Enterprise Campus Architecture, adopting a hierarchical model to improve scalability, simplify management and provide high availability.

The network follows the Cisco Enterprise Campus hierarchical model composed of:

Layer	Function
Core	High-speed backbone and inter-network routing
Distribution	Policy enforcement and traffic aggregation
Access	End-device connectivity

Table 1. Hierarchical Network Architecture Adopted by Atlas Vitae University Hospital.

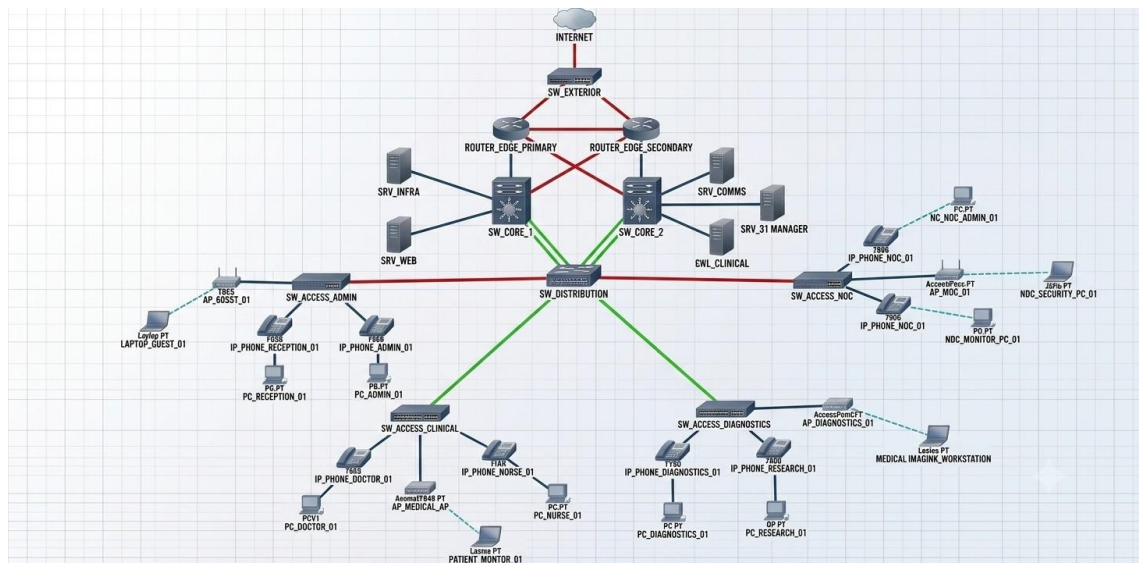


Figure 1. Logical Network Topology.

3.2 Routing

Dynamic routing is implemented using the Open Shortest Path First (OSPF) protocol operating within Area 0.

OSPF provides automatic route exchange between Layer 3 devices, allowing the network to adapt efficiently to topology changes while maintaining optimal routing paths.

Default Internet connectivity is provided through the enterprise edge routers.

3.3 High Availability

To improve service availability and minimize network downtime, the following redundancy mechanisms were implemented:

- Hot Standby Router Protocol (HSRP);
- EtherChannel using LACP;
- Spanning Tree Protocol (STP);
- Dual Enterprise Edge Routers;
- Dual Layer 3 Core Switches.

Spanning Tree Protocol was configured to provide a predictable and resilient Layer 2 topology. Core 1 was configured as the primary root bridge, while Core 2 was configured as the secondary root bridge for the enterprise VLANs. This establishes a controlled Layer 2 forwarding topology while maintaining a defined failover path if the primary root becomes unavailable.

These technologies increase network resilience and ensure continuous service availability during equipment or link failures.

4 Physical Infrastructure

4.1 Infrastructure Overview

The physical infrastructure of Atlas Vitae University Hospital was designed following enterprise networking principles, providing a reliable, scalable and resilient environment capable of supporting both clinical and administrative operations.

The network equipment is distributed throughout a four-floor hospital building and interconnected through a hierarchical architecture composed of Core, Distribution and Access layers. This design simplifies network management, improves fault isolation and facilitates future expansion.

4.2 Building Distribution

The hospital is organized into four operational floors, each dedicated to specific functional areas while remaining connected to the centralized network infrastructure.

Floor	Primary Function
Floor 0	NOC and Core Infrastructure
Floor 1	Administrative Department
Floor 2	Clinical Departments
Floor 3	Diagnostic Services and Wireless Coverage

Table 2. Building Floor Distribution

4.3 Network Infrastructure Equipment

The implemented infrastructure consists of Cisco enterprise networking equipment responsible for switching, routing, wireless connectivity and high-availability services.

Equipment	Quantity
Layer 3 Core Switches	2
Distribution Switch	1
Access Switches	4
Enterprise Edge Routers	2
Wireless Access Points	4

Table 3. Network Infrastructure Equipment

4.4 End Devices

The network supports multiple categories of end devices distributed throughout the hospital according to departmental requirements.

Device	Quantity
Desktop Computers	8
Enterprise Servers	5
Cisco IP Phones	8
Wireless Laptops	4
Network Printers	4

Table 4. Connected End Devices

4.5 Physical Infrastructure Summary

The implemented physical infrastructure provides the foundation required to support the hospital's enterprise network, ensuring connectivity between users, servers and critical services while maintaining scalability and operational reliability. The hierarchical distribution of network devices also facilitates future maintenance and infrastructure expansion.

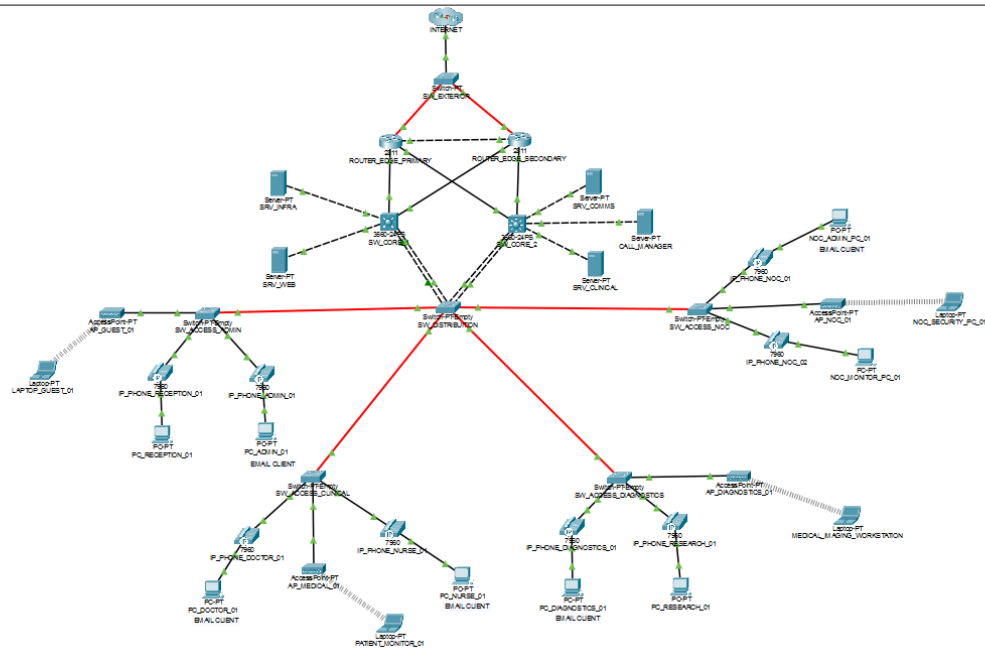


Figure 2. Physical Network Infrastructure.

Figure 2 illustrates the complete physical implementation of the enterprise network, including the Core Layer, Distribution Layer, Access Layer, Enterprise Edge Routers, servers, wireless infrastructure, IP telephony devices and end-user workstations. The topology also includes the simulated Internet environment used to validate external connectivity, NAT/PAT operation and public service accessibility.

5 Logical Infrastructure

5.1 Overview

The logical infrastructure of Atlas Vitae University Hospital was designed to provide a secure, scalable and highly available communication environment capable of supporting all hospital departments and critical healthcare services.

Unlike the physical infrastructure, which represents the hardware and physical connections, the logical infrastructure defines how devices communicate across the enterprise network. This includes network segmentation, routing, addressing, enterprise services and communication policies.

The network is logically divided into multiple Virtual Local Area Networks (VLANs), each assigned to a specific department or operational function. This segmentation improves security, reduces unnecessary broadcast traffic, simplifies network administration and allows independent management of different organizational units.

Communication between VLANs is performed at the Layer 3 Core through Switch Virtual Interfaces (SVIs), providing efficient routing while maintaining logical isolation between departments.

5.2 Enterprise Logical Design Principles

The logical architecture follows modern Cisco Enterprise Campus design principles to ensure operational continuity, security and future scalability.

5.2.1 Network Segmentation

The infrastructure is divided into dedicated VLANs for administration, medical staff, nursing, diagnostics, enterprise servers, Network Operations Center (NOC), guest wireless access, IP telephony and infrastructure management. Each VLAN operates as an independent broadcast domain, improving network organization and performance.

5.2.2 Centralized Services

Critical enterprise services such as DHCP, DNS, FTP, Web Services, Clinical Applications and Network Monitoring are centralized within the Server VLAN. This approach simplifies maintenance, strengthens security and provides consistent service availability across the entire hospital network.

5.2.3 Layer 3 Routing

Inter-VLAN communication is performed by the Core Switches, eliminating the need for router-on-a-stick solutions and providing high-performance packet forwarding throughout the campus.

5.2.4 Scalability

The logical design allows future expansion through the addition of new VLANs, subnets, departments and enterprise services without requiring major modifications to the existing infrastructure.

5.3 Inter-VLAN Communication

Each VLAN is associated with its own Switch Virtual Interface (SVI), which acts as the default gateway for devices connected to that network segment.

Whenever communication between departments is required, packets are forwarded to the corresponding SVI, where the Core Switch performs Layer 3 routing based on its routing table before delivering traffic to the destination VLAN.

This architecture provides fast communication between departments while preserving logical separation and allowing security policies to be enforced whenever necessary.

5.4 Enterprise Traffic Flow

Normal communication inside the hospital follows a hierarchical forwarding model.

User devices first communicate with their local Access Switch. Traffic is then forwarded to the Distribution Switch before reaching the Core Layer, where routing decisions are performed.

Internal communications between departments remain inside the campus network, while requests destined for centralized services are forwarded to the Server VLAN. Internet-bound traffic is routed towards the Enterprise Edge Routers, where NAT/PAT translation is performed before leaving the hospital network.

5.5 Logical Infrastructure Summary

The implemented logical infrastructure provides secure and efficient communication between every department of Atlas Vitae University Hospital. Through VLAN segmentation, Layer 3 routing, centralized enterprise services and dynamic routing protocols, the network achieves high performance, simplified administration, improved security and sufficient scalability to support future infrastructure growth

6 IP Addressing Plan

6.1 Addressing Strategy

A structured private IPv4 addressing scheme was implemented to provide consistency, simplify network administration and facilitate future expansion. Each department was assigned an independent subnet, allowing clear separation between network segments while supporting efficient Layer 3 routing.

The addressing plan follows a predictable structure, making network identification, troubleshooting and documentation considerably easier. Every VLAN has its own dedicated subnet and gateway address, which are configured on the corresponding Switch Virtual Interface (SVI) located on the Layer 3 Core Switches.

6.2 VLAN Address Allocation

The following IPv4 addressing scheme was implemented throughout the hospital infrastructure.

VLAN	Department	Network Address	Default Gateway	Subnet Mask
10	Administration	10.0.10.0	10.0.10.1	255.255.255.0 (/24)
20	Doctors	10.0.20.0	10.0.20.1	255.255.255.0 (/24)

30	Nurses	10.0.30.0	10.0.30.1	255.255.255.0 (/24)
40	Diagnostics	10.0.40.0	10.0.40.1	255.255.255.0 (/24)
50	Servers	10.0.50.0	10.0.50.1	255.255.255.0 (/24)
60	NOC	10.0.60.0	10.0.60.1	255.255.255.0 (/24)
70	Guest Network	10.0.70.0	10.0.70.1	255.255.255.0 (/24)
80	Voice	10.0.80.0	10.0.80.1	255.255.255.0 (/24)
90	Management	10.0.90.0	10.0.90.1	255.255.255.0 (/24)

Table 5. IPv4 Addressing Plan

6.3 Infrastructure Networks

In addition to user VLANs, dedicated point-to-point networks were implemented to interconnect the routing infrastructure.

These networks are used exclusively for communication between the Core Switches and Enterprise Edge Routers, as well as for router-to-router connectivity required by OSPF and HSRP operation.

Purpose	Network
Core 1 ↔ Edge Primary	10.255.0.0/30
Edge Primary ↔ Edge Secondary	10.255.0.8/30
Edge Secondary ↔ Core 2	10.255.0.4/30
Core 1 ↔ Edge Secondary (redundant link)	10.255.0.12/30
Core 2 ↔ Edge Primary (redundant link)	10.255.0.16/30

Table 6. Infrastructure Point-to-Point Networks

6.4 IP Address Assignment

Client devices obtain their IPv4 configuration dynamically through the centralized DHCP Server located in the Server VLAN. Static addressing is reserved for infrastructure devices, including routers, switches, servers, printers and network management systems.

This combination of dynamic and static addressing simplifies network administration while ensuring that critical infrastructure components always maintain predictable IP addresses.

6.5 Addressing Plan Summary

The implemented addressing plan provides a structured, scalable and easily maintainable IPv4 infrastructure. The separation of departmental networks into dedicated subnets simplifies routing, enhances security and allows future expansion without requiring major modifications to the existing addressing scheme.

7 VLAN Design

7.1 VLAN Segmentation Strategy

Virtual Local Area Networks (VLANs) were implemented to logically separate the hospital's departments and services into independent broadcast domains. This approach improves network performance, enhances security and simplifies administration by isolating traffic according to organizational functions.

Each VLAN was created to support a specific operational area, allowing access policies to be applied according to the role of its users and devices. This segmentation also reduces broadcast traffic and limits the impact of network failures to individual network segments.

7.2 VLAN Allocation

The following VLAN structure was implemented throughout the enterprise network.

VLAN ID	VLAN Name	Purpose
10	ADMIN	Administrative Department
20	DOCTORS	Medical Staff
30	NURSES	Nursing Department
40	DIAGNOSTICS	Diagnostic Equipment and Imaging Systems

50	SERVERS	Enterprise Servers and Centralized Services
60	NOC	Network Operations Center
70	GUEST	Guest Wireless Network
80	VOICE	Cisco IP Telephony Infrastructure
90	MANAGEMENT	Network Infrastructure Management
99	NATIVE	Native VLAN and Unused Switch Ports

Table 7. VLAN Allocation

7.3 Trunk Infrastructure

Inter-switch communication is performed through IEEE 802.1Q trunk links, allowing multiple VLANs to traverse the same physical connection.

All trunk links transport the required VLANs while maintaining VLAN 99 as the native VLAN. This configuration follows Cisco best practices by separating user traffic from native trunk traffic.

Unused switch interfaces were assigned to VLAN 99 and administratively disabled to reduce the attack surface of the network.

7.4 EtherChannel Implementation

To increase bandwidth and improve fault tolerance, Link Aggregation Control Protocol (LACP) EtherChannel was implemented between the Core Layer and the Distribution Layer.

By combining multiple physical interfaces into a single logical link, EtherChannel provides load balancing and redundancy. In the event of a physical link failure, communication continues through the remaining active interfaces without interrupting network operation.

This implementation increases both network performance and infrastructure availability.

7.5 Voice and Management VLANs

A dedicated Voice VLAN (VLAN 80) was implemented to support Cisco IP Telephony devices. Separating voice traffic from data traffic improves network organization and facilitates the application of Quality of Service (QoS) policies.

Infrastructure devices, including switches and network administrators' workstations, operate within the Management VLAN (VLAN 90). Isolating management traffic from user networks improves security and simplifies infrastructure administration.

7.6 VLAN Design Summary

The implemented VLAN architecture provides efficient logical segmentation, reduces broadcast domains and establishes a secure communication environment for all hospital departments. Combined with Inter-VLAN Routing, EtherChannel and dedicated Voice and Management VLANs, the design offers a scalable and resilient foundation capable of supporting current operational requirements as well as future infrastructure expansion.

8 Enterprise Network Services (and technologies)

8.1 Overview

Atlas Vitae University Hospital provides a centralized set of enterprise network services responsible for supporting daily clinical, administrative and operational activities.

All critical services are deployed within the Server VLAN whenever possible, allowing centralized management, simplified maintenance and improved security. Additional infrastructure services such as Cisco Call Manager Express, QoS and HSRP are implemented directly on the network infrastructure devices where required.

The implemented services provide automatic network configuration, name resolution, web access, file transfer, electronic mail, IP telephony, network monitoring, time synchronization, centralized logging and traffic prioritization.

8.2 Enterprise Services

Service	Purpose	Status
DHCP	Automatic IP Address Assignment	Operational
DNS	Hostname Resolution	Operational
Internal Web Server	Internal Hospital Applications	Operational
Public Web Server	Public Patient Portal	Operational
FTP Server	File Transfer	Operational
Email Server	Enterprise Communications	Operational
Cisco Call Manager Express	IP Telephony	Operational
Network Monitoring Server	Infrastructure Monitoring	Operational

NTP	Time Synchronization	Operational
Syslog	Centralized Log Collection	Operational

Table 8. Enterprise Network Services

8.2.1 DHCP Service

Dynamic Host Configuration Protocol (DHCP) automatically assigns IPv4 configuration to client devices. The centralized DHCP server distributes IP addresses, subnet masks, default gateways and DNS server information. DHCP Relay (ip helper-address) is configured on the Core Switches to forward DHCP requests from all client VLANs.

8.2.2 DNS Service

The Domain Name System (DNS) provides centralized hostname resolution for both internal and public services. Users access hospital resources through domain names instead of IP addresses, simplifying resource identification and management.

8.2.3 Web Services

Three independent web platforms were implemented to support different operational requirements within the hospital network.

- **Internal Web Server:** Hosts internal hospital applications and administrative resources accessible only to authorized internal users.
- **Public Web Server:** Publishes public hospital information and patient services through Static NAT, allowing secure access from the Internet.
- **Network Monitoring Platform (NOC Dashboard):** Provides a centralized web-based interface for monitoring the enterprise infrastructure, including network devices, servers, services, security status and overall network performance. For security reasons, access to this platform is restricted exclusively to Network Operations Center (NOC) administrators located in **VLAN 60**.

Access to the public server is protected by Access Control Lists configured on the Enterprise Edge Router.

8.2.4 FTP Service

The FTP server provides centralized file storage and controlled file transfer between authorized departments. It supports software distribution, backup procedures and administrative document exchange.

8.2.5 Email Service

The Email Server supports internal electronic communications between hospital departments, allowing secure exchange of operational information across the enterprise network.

8.2.6 Cisco IP Telephony

Voice communications are provided through Cisco Call Manager Express (CME), configured on the Enterprise Edge Primary Router.

Cisco IP Phones operate inside the dedicated Voice VLAN (VLAN 80) and automatically receive their configuration through DHCP before registering with the Call Manager Express service.

8.2.7 Network Time Protocol (NTP)

Network Time Protocol (NTP) was configured to synchronize the clocks of routers, switches and network devices. Consistent timestamps improve event correlation, simplify troubleshooting and ensure accurate logging across the entire infrastructure.

8.2.8 Syslog

A centralized Syslog server collects event messages generated by the network infrastructure. System logs include interface status changes, routing events, security notifications and device warnings, providing administrators with a centralized platform for diagnostics and auditing.

8.2.9 Quality of Service (QoS)

Quality of Service (QoS) was implemented to prioritize delay-sensitive traffic across the enterprise network.

Voice traffic generated by the Cisco IP Telephony infrastructure receives higher transmission priority than standard data traffic, reducing latency, jitter and packet loss during periods of network congestion.

This prioritization ensures reliable voice communications while maintaining efficient delivery of normal data traffic.

8.3 Enterprise Services Summary

The combination of centralized network services and enterprise infrastructure technologies provides a reliable, secure and highly available communication environment. DHCP, DNS, Web Services, FTP, Email, Call Manager Express, Network Monitoring, NTP, Syslog and QoS collectively support the operational requirements of Atlas Vitae University Hospital while ensuring scalability, maintainability and continuous service availability.

9 Internet Connectivity and NAT/PAT Implementation

9.1 Internet Connectivity Overview

Atlas Vitae University Hospital was designed to operate as a private enterprise network while maintaining secure connectivity with external Internet resources.

To simulate a real-world Internet environment, an Internet Service Provider (ISP) infrastructure was implemented inside Cisco Packet Tracer. This simulated ISP network hosts external services that can be accessed by the hospital users, including:

- **www.google.com**
- **www.globalhealth.org**

These external servers represent public Internet resources and allow realistic testing of routing, DNS resolution, Network Address Translation (NAT), access control and Internet connectivity.

Communication between the enterprise network and the simulated Internet is performed through two Enterprise Edge Routers connected to the ISP infrastructure.

9.2 Public IPv4 Address Allocation

A public IPv4 address block was allocated to the hospital by the simulated Internet Service Provider.

The organization owns the public network:

200.0.0.0/26

Rather than using the entire block as a single network, it was subdivided into smaller subnets according to the intended purpose of each address range.

Public Subnet	Purpose
200.0.0.0/29	Enterprise Edge ↔ ISP Connection
200.0.0.8/29	Static NAT for Public Servers
200.0.0.16/29	PAT for VLAN 10 and VLAN 20
200.0.0.24/29	PAT for VLAN 30 and VLAN 40
200.0.0.32/29	PAT for VLAN 60 and VLAN 80
200.0.0.40/29	PAT for Guest Network (VLAN 70)
200.0.0.48/29	Reserved for Future Expansion
200.0.0.56/29	Reserved for Future Expansion

Table 9. Public IPv4 Address Allocation

This addressing strategy simplifies administration while allowing future growth without requiring address redesign.

9.3 Static Network Address Translation (Static NAT)

Static NAT was implemented to publish enterprise services that must be accessible from the Internet.

Each internal server is permanently mapped to a dedicated public IPv4 address.

This one-to-one translation allows external users to access public hospital services while preventing direct exposure of the internal addressing scheme.

The primary published service is the Public Web Server, which hosts the hospital's external website.

Static NAT ensures permanent reachability of this service regardless of changes occurring inside the private network.

9.4 Dynamic Network Address Translation (Dynamic NAT)

Dynamic NAT was implemented to provide temporary one-to-one translation between internal private addresses and addresses from a configured public NAT pool.

Unlike Static NAT, where a permanent mapping is maintained for a specific internal server, Dynamic NAT assigns an available public address from the configured pool when an eligible internal host initiates communication with an external network.

In this implementation, Dynamic NAT provides an intermediate translation mechanism for internal devices that require temporary public addressing while avoiding the need to permanently assign a public address to each host.

Once the translation is no longer required and the corresponding session expires, the public address can be returned to the NAT pool and made available for subsequent translations.

Dynamic NAT therefore complements Static NAT and PAT by providing temporary address translation for selected internal traffic while preserving the structured use of the available public IPv4 address space.

9.5 Port Address Translation (PAT)

Internal users access the Internet through Port Address Translation (PAT), also known as NAT Overload.

Instead of assigning a public IPv4 address to every internal device, multiple private hosts share the same public address by using different TCP and UDP port numbers.

To improve traffic organization, independent PAT pools were allocated for different VLAN groups.

This approach simplifies traffic identification while efficiently utilizing the available public address space.

9.6 Internet Traffic Flow

When an internal workstation accesses an Internet resource, packets follow the enterprise routing infrastructure until reaching the Enterprise Edge Routers.

The Edge Routers perform PAT translation before forwarding the packets toward the ISP network.

When responses return from the Internet, the Edge Routers use the NAT translation table to associate each session with the originating internal host before forwarding the traffic back into the enterprise network.

For services published through Static NAT, incoming Internet traffic is translated directly toward the corresponding internal server according to the configured static mappings.

9.7 High Availability

Internet connectivity is protected through two Enterprise Edge Routers operating together with Hot Standby Router Protocol (HSRP).

Under normal operating conditions, the Primary Edge Router provides Internet connectivity.

If the Primary Edge Router becomes unavailable, the Secondary Edge Router automatically assumes the virtual gateway address, minimizing service interruption for internal users.

This redundancy mechanism significantly increases the availability of Internet services.

9.8 Internet Connectivity Summary

The Internet connectivity architecture combines OSPF dynamic routing, HSRP gateway redundancy, Static NAT, Dynamic NAT and Port Address Translation (PAT) to provide secure and efficient communication between the enterprise network and external resources.

Static NAT provides permanent public mappings for services that must be reachable from the Internet. Dynamic NAT provides temporary one-to-one translations using configured public address pools, while PAT allows multiple internal hosts to share public IPv4 addresses through port-based translation.

The use of a structured public IPv4 addressing plan together with dedicated NAT mechanisms creates an enterprise-grade WAN design that supports both public-facing services and controlled Internet access for internal users.

10 Security Implementation

10.1 Overview

Security was one of the primary design objectives during the implementation of the Atlas Vitae University Hospital network.

The hospital network transports sensitive information, including administrative communications, medical services, diagnostic systems and infrastructure management traffic. Consequently, multiple security mechanisms were implemented to minimize unauthorized access while ensuring that legitimate users maintain access to the resources required for their daily activities.

Rather than relying on a single protection mechanism, the network adopts a layered security approach based on logical segmentation, access control, infrastructure protection and secure remote administration.

10.2 VLAN Segmentation

The first security layer consists of logical network segmentation through Virtual Local Area Networks (VLANs).

Each department operates inside an independent broadcast domain, preventing direct Layer 2 communication between unrelated departments.

This segmentation limits broadcast traffic, reduces the propagation of network attacks and simplifies policy enforcement across the enterprise.

Critical infrastructure such as servers, management devices, Voice over IP equipment and guest users were intentionally isolated into dedicated VLANs.

10.3 Access Control Lists (ACLs)

Access Control Lists were implemented to regulate communication between network segments according to operational requirements.

Instead of allowing unrestricted communication, only the traffic required for hospital operations is permitted.

The implemented ACLs enforce the following security policies:

- Administrative users are allowed to access enterprise services.
- Medical staff can communicate with centralized servers.
- Diagnostic equipment is restricted to the services required for healthcare operations.

- Guest users are isolated from the internal enterprise network and are only permitted Internet access.
- Voice devices communicate only with the Cisco Call Manager Express service.
- Infrastructure management is restricted exclusively to authorized network administrators.
- The Network Monitoring Dashboard is accessible only from the Network Operations Center (VLAN 60).

This policy follows the principle of **least privilege**, where users receive only the minimum level of access required to perform their duties.

10.4 Public Service Protection

Only a single internal service is published to the Internet: the Public Web Server.

Static NAT maps the internal web server to its public IPv4 address while an inbound Access Control List applied on the Enterprise Edge Router permits only HTTP and HTTPS traffic originating from the Internet.

All remaining unsolicited traffic directed toward the enterprise network is denied.

This configuration significantly reduces the attack surface exposed to external networks.

10.5 Guest Network Isolation

Wireless guest users operate exclusively within VLAN 70.

This network is completely isolated from the enterprise infrastructure through Access Control Lists configured on the Layer 3 Core Switches.

Guest devices cannot communicate with:

- Administrative workstations
- Medical departments
- Diagnostic systems
- Enterprise servers
- Infrastructure management devices
- Network monitoring systems

Their connectivity is limited exclusively to Internet access through Port Address Translation (PAT).

10.6 Secure Infrastructure Management

Infrastructure administration is performed through the dedicated Management VLAN (VLAN 90).

Only authorized administrators are permitted to access the management interfaces of switches and routers.

Remote administration is performed using **Secure Shell (SSH)**, ensuring that management traffic is encrypted while traversing the network.

The use of SSH replaces insecure remote management protocols and protects administrative credentials against interception.

10.7 Switch Port Protection

To reduce the possibility of unauthorized physical access, unused switch interfaces were assigned to the Native VLAN (VLAN 99) and administratively disabled.

This prevents unauthorized devices from being connected to unused network ports and follows Cisco enterprise security best practices.

10.8 Layered Security Approach

The implemented security architecture combines multiple independent protection mechanisms, including:

- VLAN segmentation
- Layer 3 Access Control Lists
- Guest network isolation
- Infrastructure management separation
- Secure remote administration using SSH
- Static NAT protection for public services
- Disabled unused switch ports
- Dedicated Management and Voice VLANs

Together, these mechanisms provide defense in depth, ensuring that the compromise of one security layer does not automatically expose the entire enterprise network.

10.9 Security Summary

The implemented security architecture protects both users and infrastructure through logical segmentation, controlled access policies and secure administrative practices. By combining VLAN isolation, ACL enforcement, secure management protocols and controlled Internet exposure, the Atlas Vitae University Hospital network achieves a security model consistent with modern enterprise networking practices while maintaining operational efficiency and service availability.

11 Network Validation

11.1 Validation Overview

Following the implementation phase, a comprehensive validation process was performed to verify the correct operation of the enterprise network infrastructure.

Each major component of the network was individually tested to ensure that connectivity, routing, redundancy, security policies and enterprise services operated according to the project requirements.

11.1.1 OSPF Adjacency Validation

Objective

Verify that all OSPF neighbors successfully established adjacencies.

Validation Command

show ip ospf neighbor

Expected Result

All expected neighbors are in the **FULL** state.

```
EDGE_PRIMARY#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/DR	00:00:39	10.255.0.10	FastEthernet0/1
1.1.1.1	1	FULL/BDR	00:00:32	10.255.0.2	FastEthernet0/0

```
EDGE_PRIMARY#
```

Figure 3. OSPF Neighbor Table

11.1.2 HSRP Validation

Objective

Verify the operation of gateway redundancy.

Validation Commands

show standby brief

```
EDGE_PRIMARY#show standby brief
```

P indicates configured to preempt.

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fal/0	10	120	P	Active	local	200.0.0.3	200.0.0.1

```
EDGE_PRIMARY#
```

Figure 4. HSRP Operational State

11.1.3 EtherChannel Validation

Validation Commands

show etherchannel summary

Expected Result

Port-channel status SU.

```
SW_DISTRIBUTION#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  S - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 2
Number of aggregators:          2

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1 (SU)      LACP        Gig0/1 (P) Gig1/1 (P)
2      Po2 (SU)      LACP        Gig2/1 (P) Gig3/1 (P)
SW_DISTRIBUTION#
```

Figure 5. EtherChannel Summary

11.1.4 Spanning Tree Validation

```
CORE_1#show spanning-tree vlan 60
VLAN0060
  Spanning tree enabled protocol ieee
  Root ID    Priority    24636
             Address     00D0.5844.DA10
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24636 (priority 24576 sys-id-ext 60)
             Address     00D0.5844.DA10
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----+-----+-----+-----
Po1                      Desg FWD 3         128.27 P2p
```

Figure 6. Spanning-Tree Validation.

11.1.5 VLAN Validation

```
CORE_1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	
10	ADMIN	active	
20	DOCTORS	active	
30	NURSES	active	
40	DIAGNOSTICS	active	
50	SERVERS	active	Fa0/3, Fa0/4
60	NOC	active	
70	GUEST	active	
80	VOICE	active	
90	MANAGEMENT	active	
99	NATIVE	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
CORE_1#
```

Figure 7. VLAN Validation

11.1.6 DHCP Validation

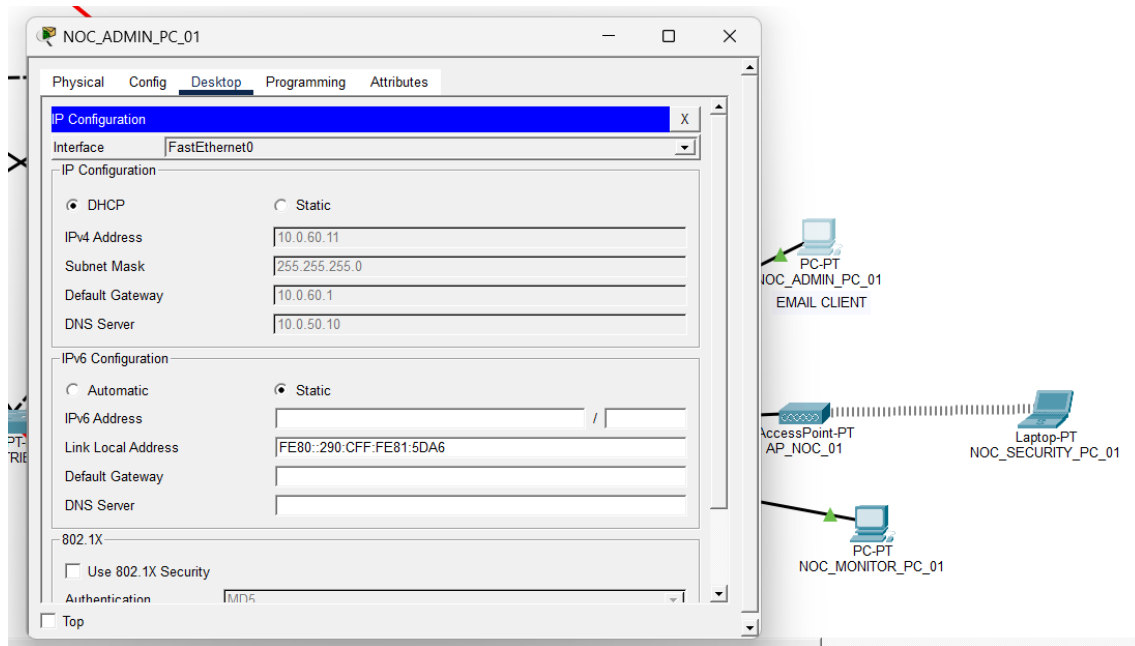


Figure 8. DHCP Address Assignment

11.1.7 Internet Connectivity

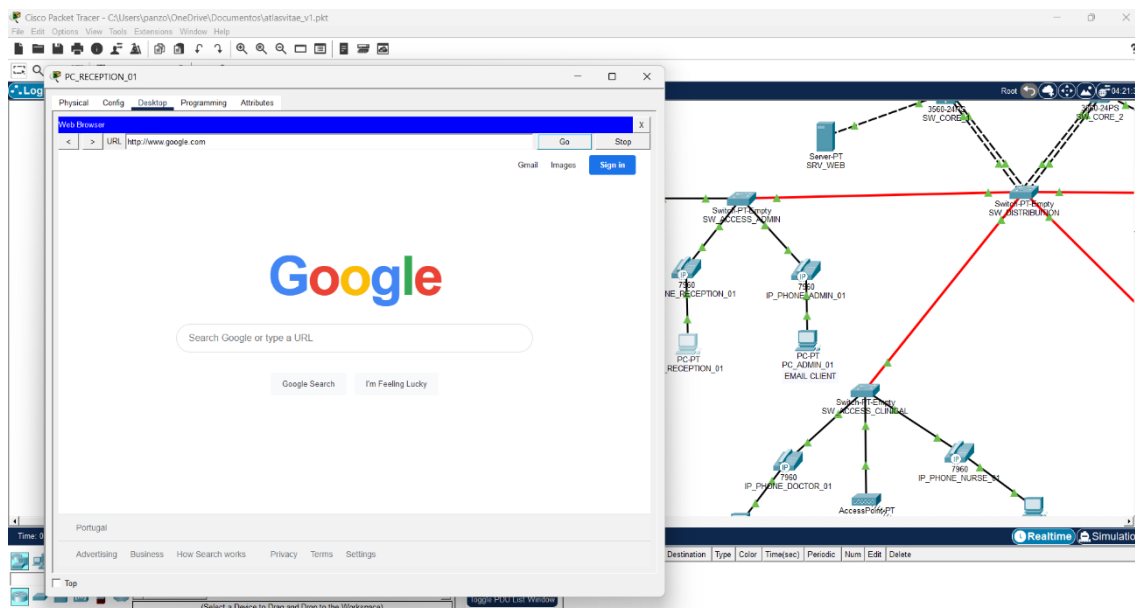


Figure 9. Internet Connectivity Validation 1.

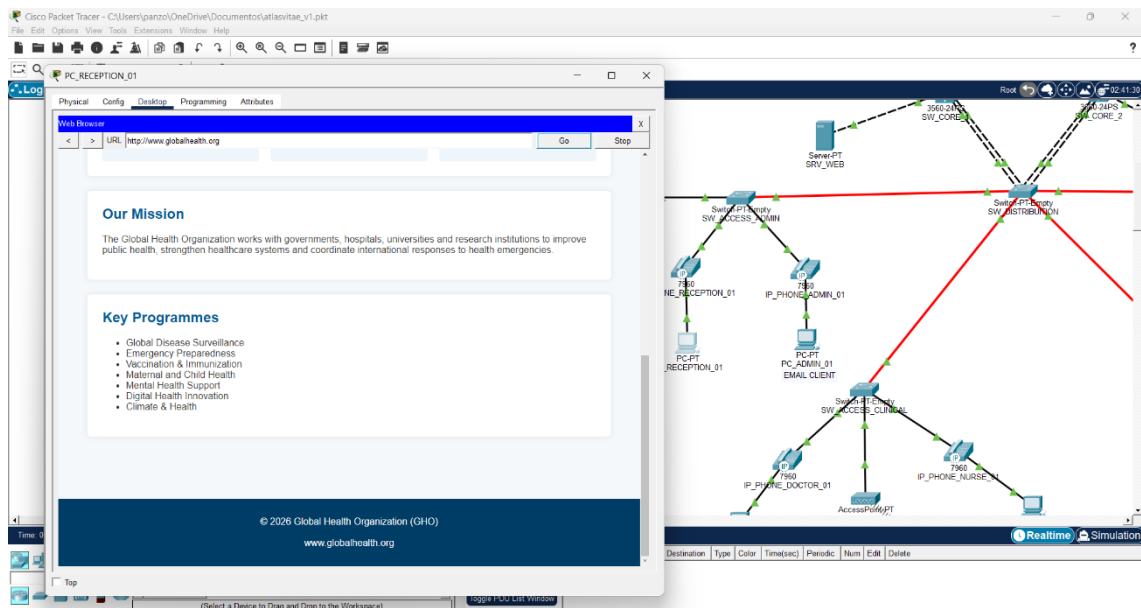


Figure 10. Internet Connectivity Validation 2.

11.1.8 Public Web Server

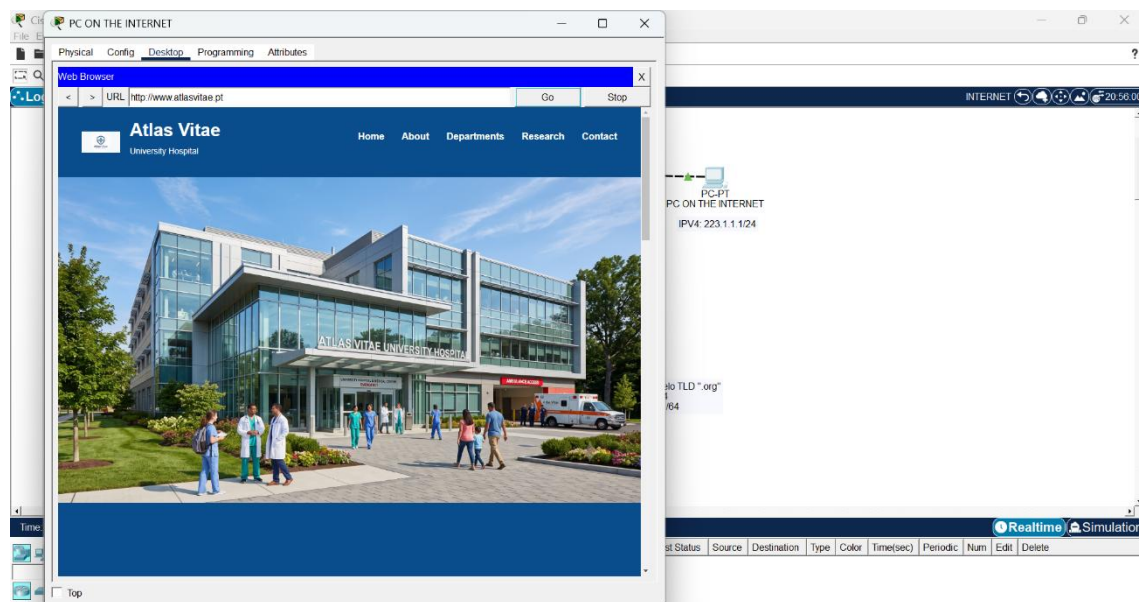


Figure 11. Public Website Access.

11.1.9 Internal Web Server

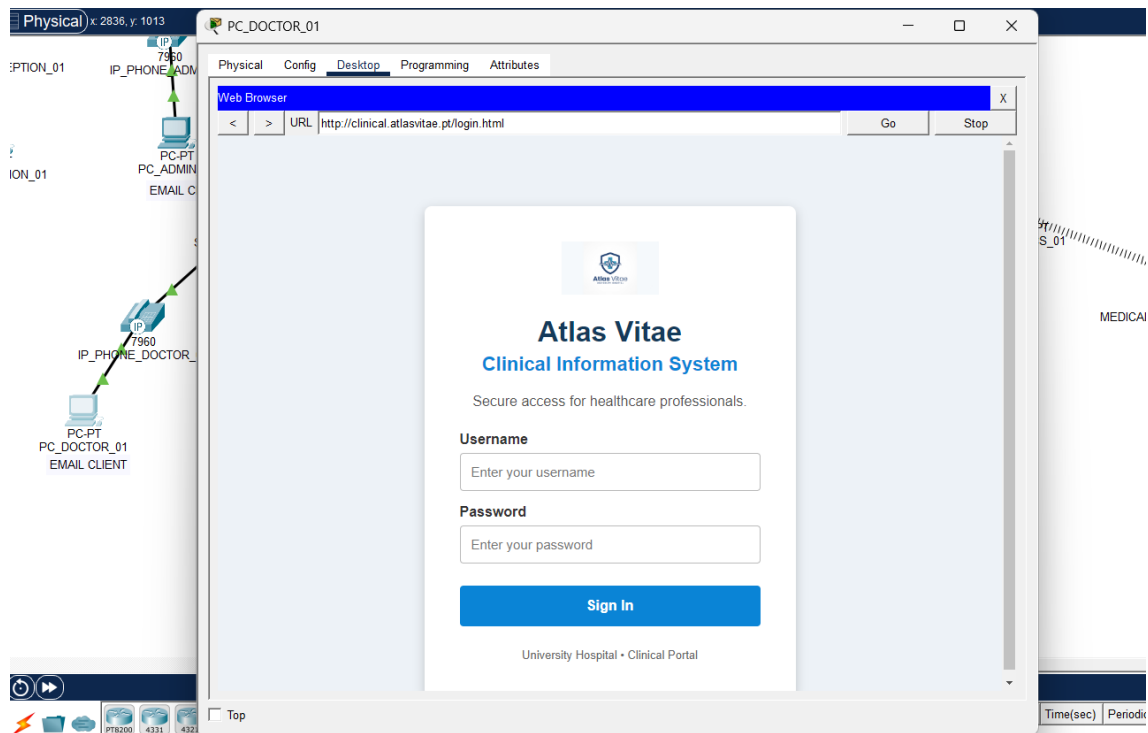


Figure 12. Clinical Website Validation

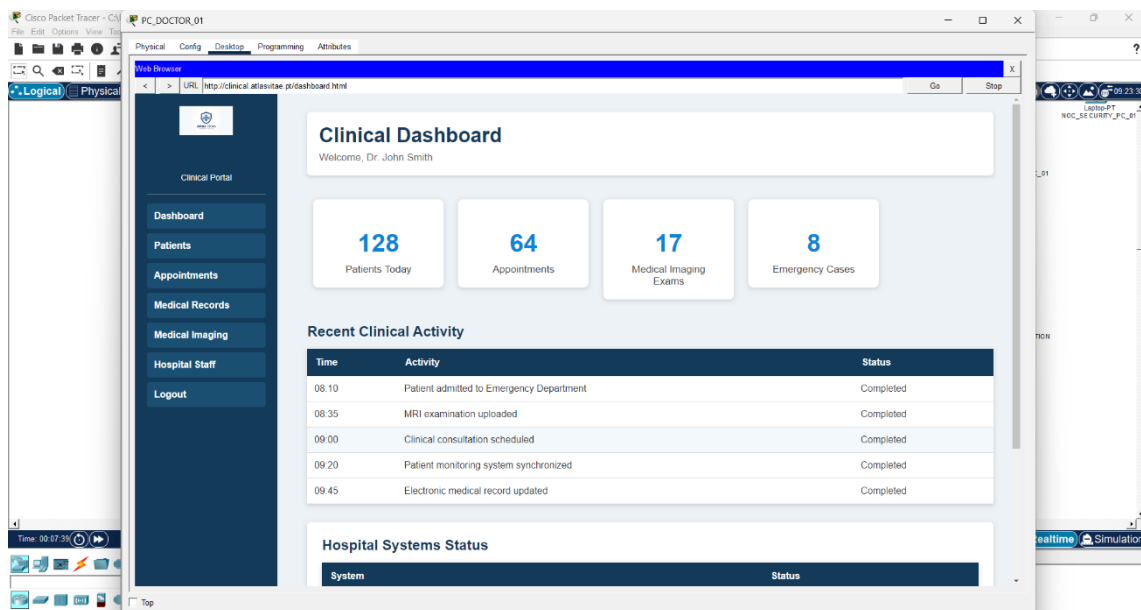


Figure 13. Clinical site Dashboard.

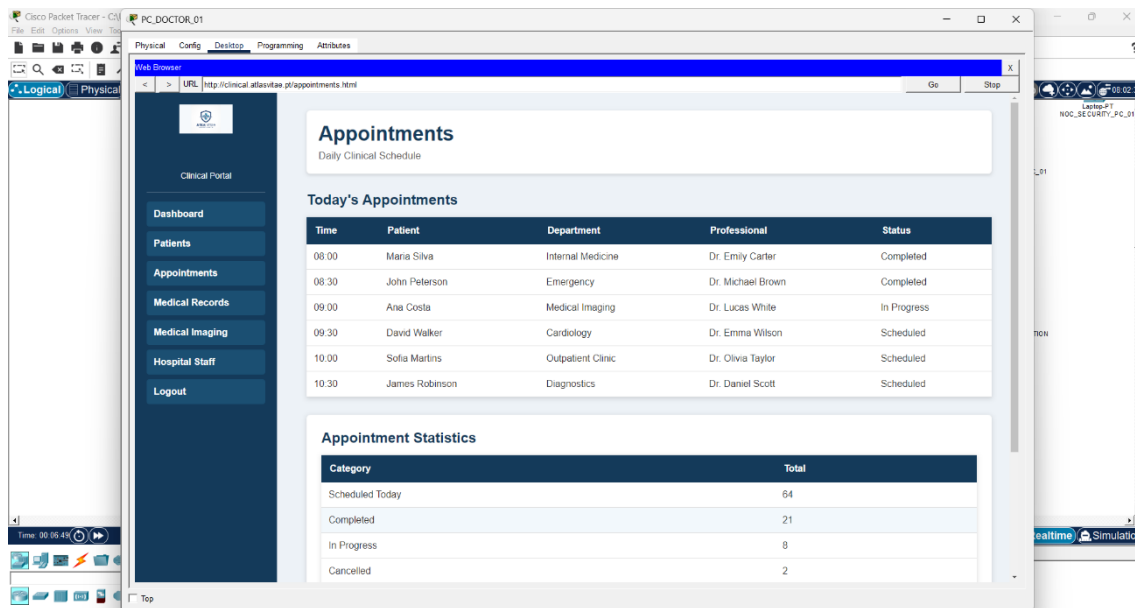


Figure 14. Clinical site appointments.

11.1.10 Network Monitoring Site

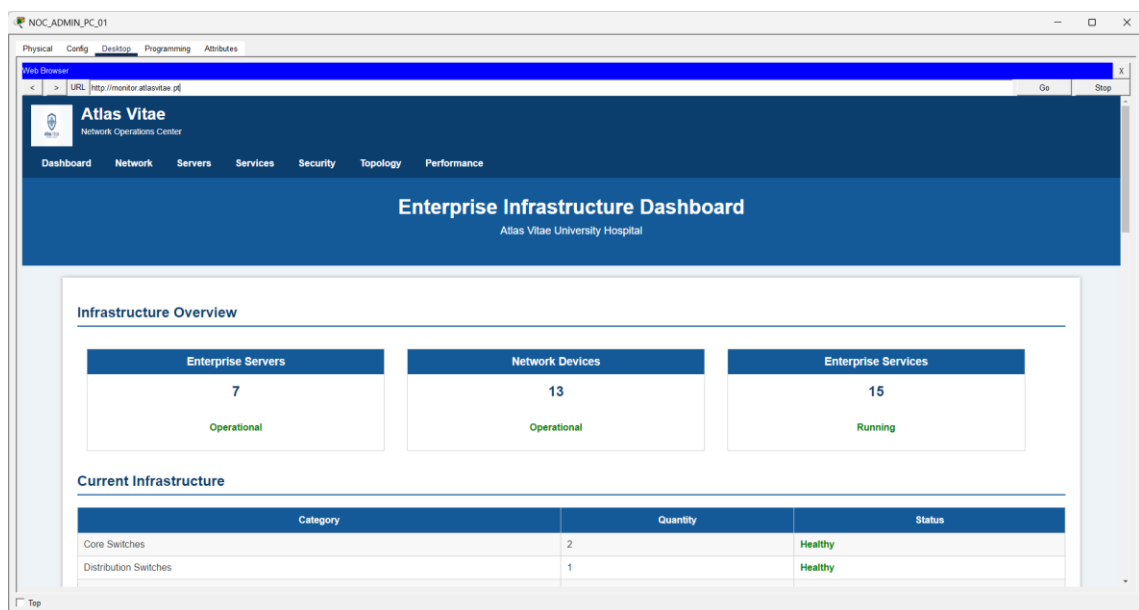


Figure 15. Monitor site Dashboard.

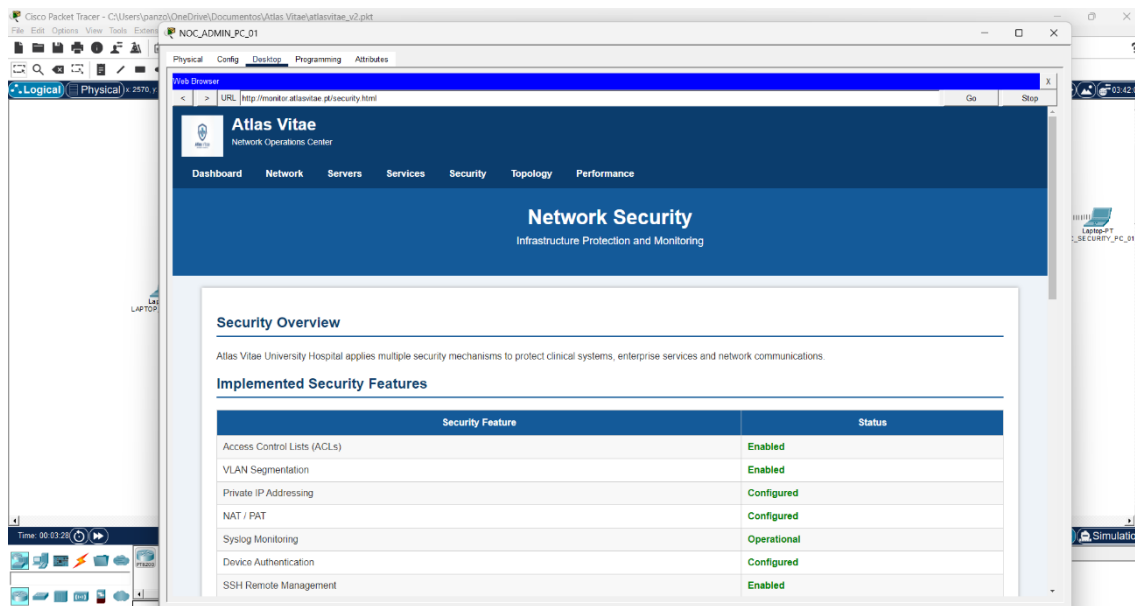


Figure 16. Monitor site Security.

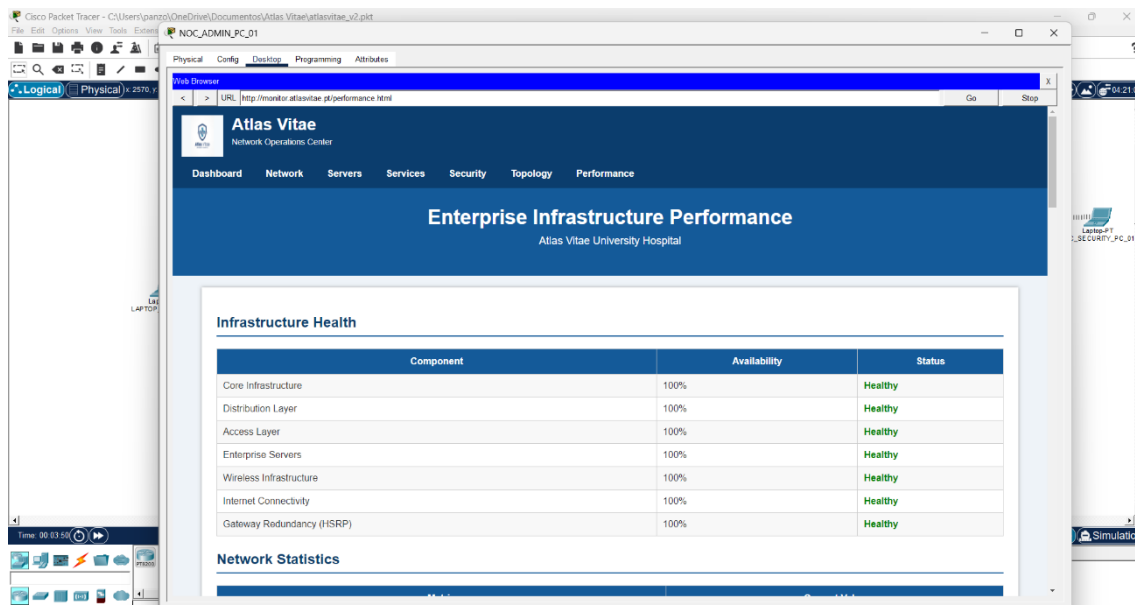


Figure 17. monitor site performance.

11.1.11 ACL Validation

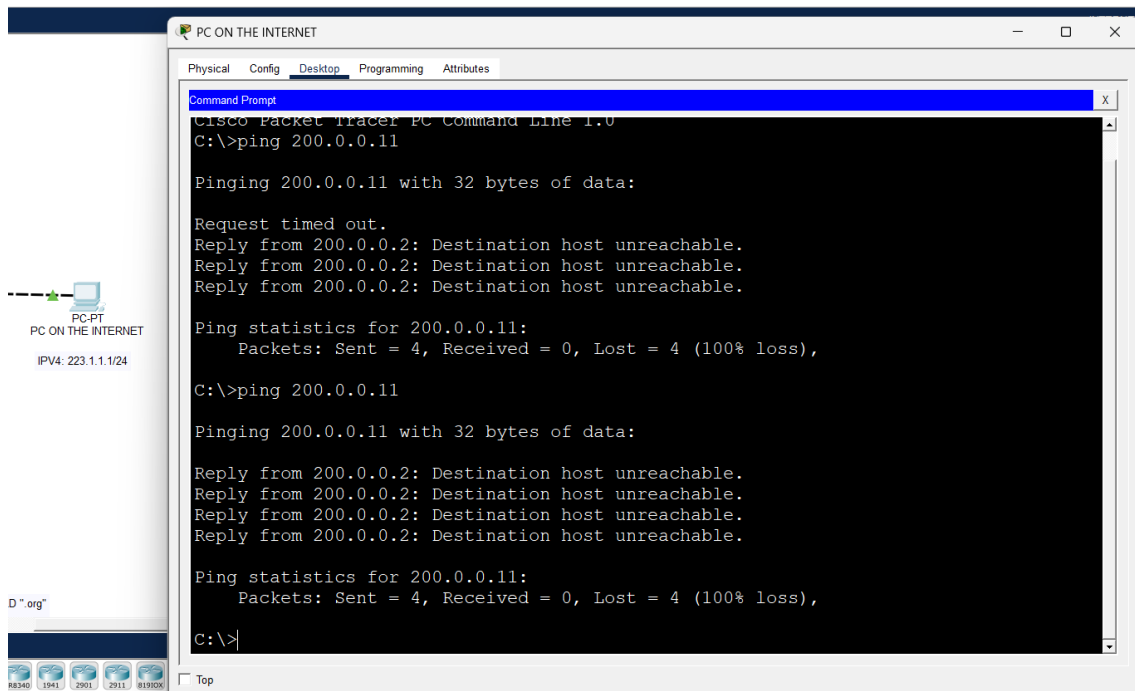


Figure 18. ACL Validation 1, The Internet PC cannot ping the company's internal end devices.

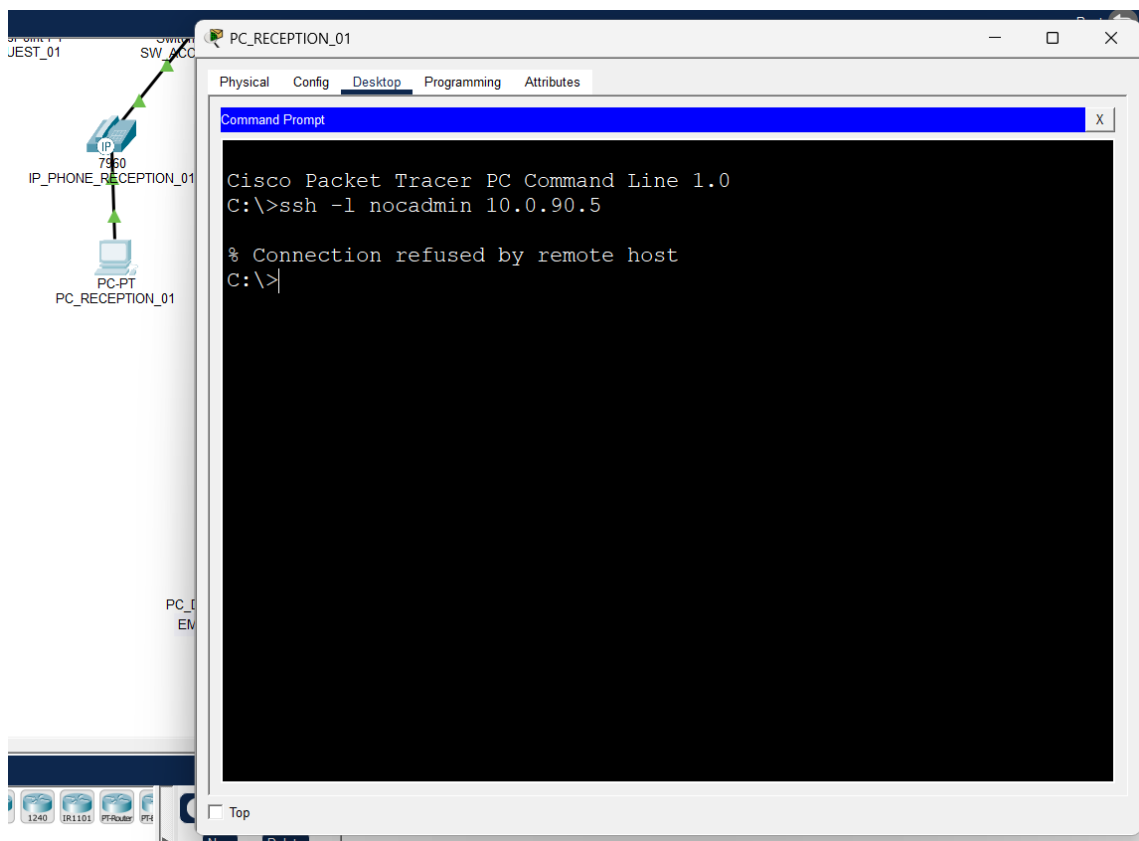


Figure 19. ACL Validation 2, No company PC, except NOCAdmin, can remotely access the network devices via SSH.

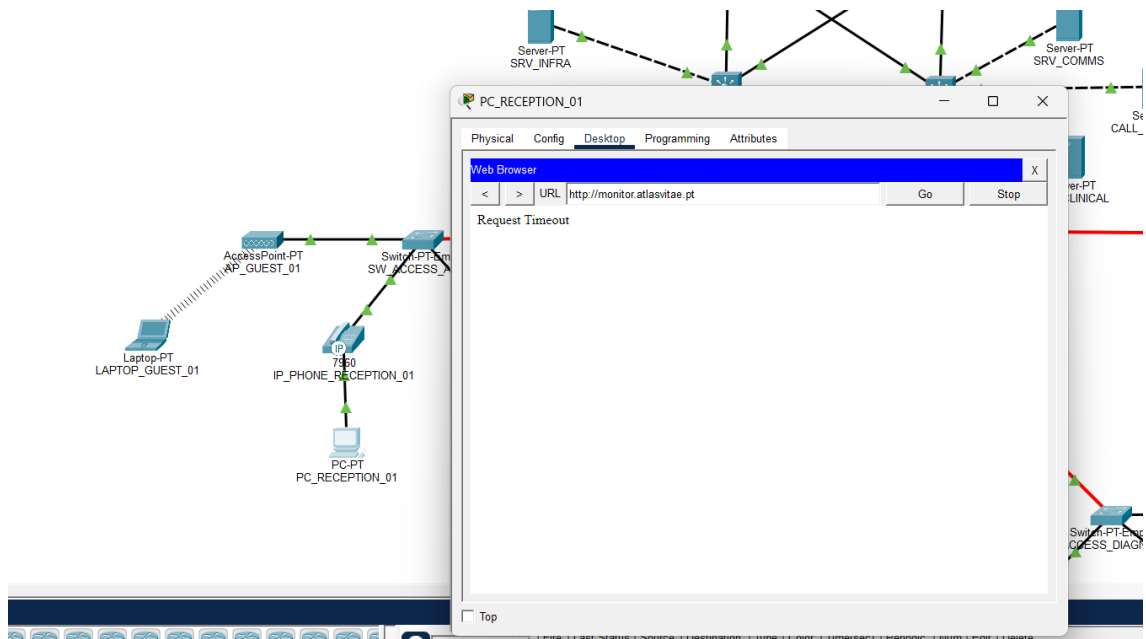


Figure 20. ACL Validation 3, PCs from any other VLAN cannot access the monitoring website; access is restricted to PCs in VLAN 60 (NOC).

11.1.12 SSH Validation

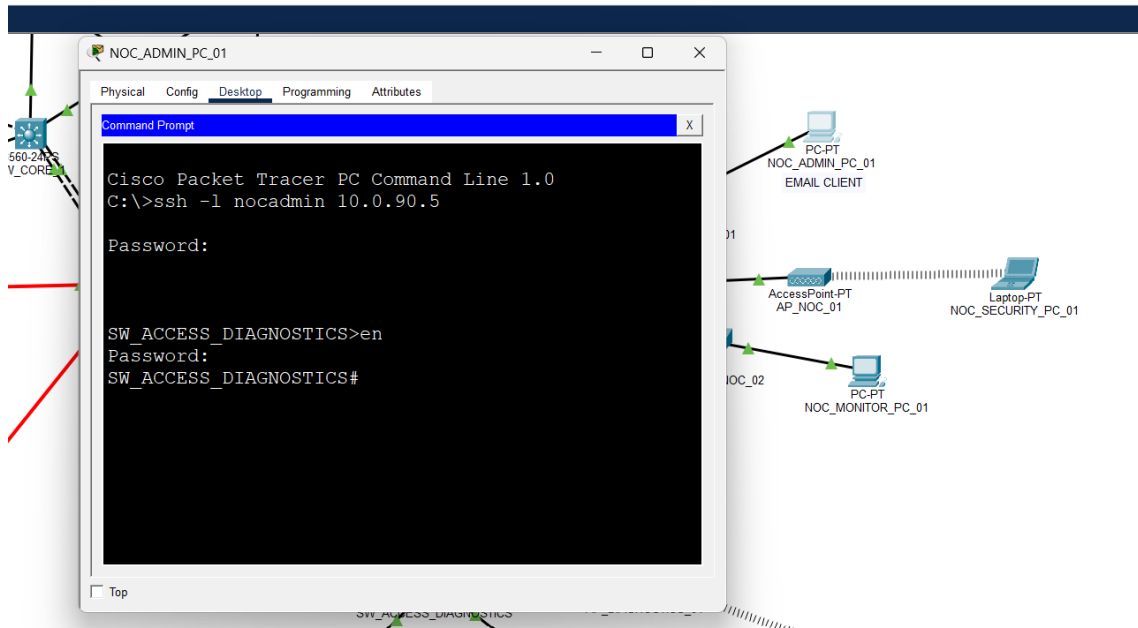


Figure 21. SSH Remote Access.

11.1.13 QoS Validation

```
EDGE_PRIMARY#show policy-map
Policy Map HOSPITAL_QOS
  Class CLINICAL
    Bandwidth 25 (kbps) Max Threshold 64 (packets)
  Class ADMIN
    Bandwidth 15 (kbps) Max Threshold 64 (packets)
  Class VOICE
    Strict Priority
    Bandwidth 30 (kbps) Burst 750 (Bytes)
  Class class-default
    Flow based Fair Queueing
    Bandwidth 0 (kbps) Max Threshold 64 (packets)
EDGE_PRIMARY#show class-map
Class Map match-any class-default (id 0)
  Match any
Class Map match-any CLINICAL (id 1)
  Match access-group name ACL_CLINICAL
Class Map match-any ADMIN (id 2)
  Match access-group name ACL_ADMIN
Class Map match-any VOICE (id 3)
  Match access-group name ACL_VOICE
EDGE_PRIMARY#
```

Figura 22. Qos validation.

11.1.14 IP Telephony



Figure 23. Cisco IP Telephony Validation.

11.1.15 FTP Service Validation

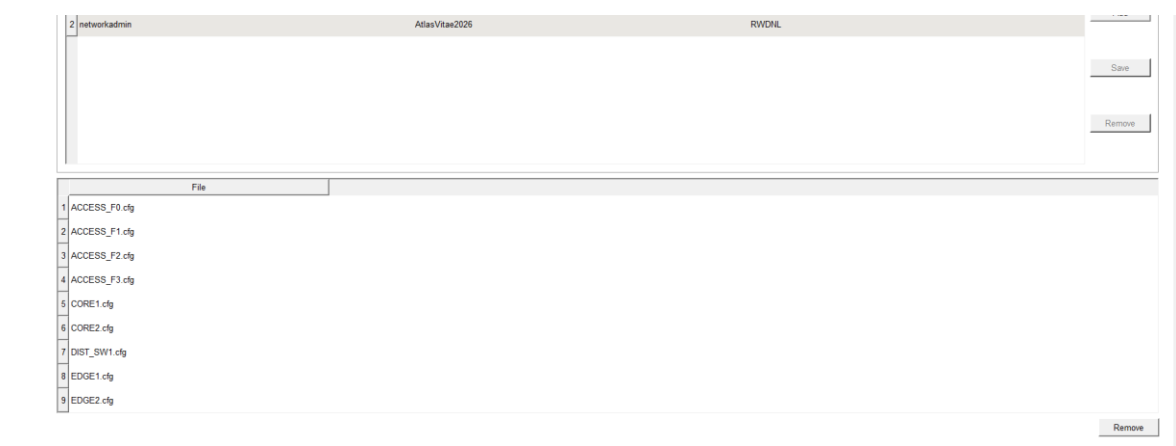


Figure 24. FTP File Transfer Validation.

11.1.16 Syslog Service Validation

Syslog			
Service			
	Time	Hostname	Message
1	07.20.2026 11:53:47.694 AM	10.255.0.13	%HSRP-6-STATECHANGE: FastEthernet1/0 Grp 10 state Speak -> Standby
2	07.20.2026 11:53:51.118 AM	10.0.50.1	11.53.51: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/2 from LOADING ...
3	07.20.2026 11:53:51.113 AM	10.255.0.13	11.53.51: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/1 from LOADING ...
4	07.20.2026 11:53:56.260 AM	10.255.0.1	%IPPHONE-6-REGISTER: ephone-5 IP: 10.0.80.16 Socket: 2 DeviceType Phone has ...
5	07.20.2026 11:53:56.341 AM	10.255.0.1	%IPPHONE-6-REGISTER: ephone-4 IP: 10.0.80.15 Socket: 2 DeviceType Phone has ...
6	07.20.2026 11:53:56.375 AM	10.255.0.1	%IPPHONE-6-REGISTER: ephone-1 IP: 10.0.80.14 Socket: 2 DeviceType Phone has ...
7	07.20.2026 11:53:56.462 AM	10.255.0.1	%IPPHONE-6-REGISTER: ephone-3 IP: 10.0.80.13 Socket: 2 DeviceType Phone has ...
8	07.20.2026 11:53:56.547 AM	10.255.0.1	%IPPHONE-6-REGISTER: ephone-6 IP: 10.0.80.17 Socket: 2 DeviceType Phone has ...
9	07.20.2026 11:53:57.001 AM	10.255.0.1	%IPPHONE-6-REGISTER: ephone-2 IP: 10.0.80.11 Socket: 2 DeviceType Phone has ...
10	07.20.2026 11:53:57.120 AM	10.255.0.1	%IPPHONE-6-REGISTER: ephone-7 IP: 10.0.80.12 Socket: 2 DeviceType Phone has ...
11	07.20.2026 11:53:57.145 AM	10.255.0.1	%IPPHONE-6-REGISTER: ephone-8 IP: 10.0.80.10 Socket: 2 DeviceType Phone has ...
12	07.20.2026 11:56:03.986 AM	10.255.0.13	%HSRP-6-STATECHANGE: FastEthernet1/0 Grp 10 state Standby -> Active
13	07.20.2026 11:56:13.341 AM	10.255.0.13	%HSRP-6-STATECHANGE: FastEthernet1/0 Grp 10 state Speak -> Standby
14	07.20.2026 11:58:06.560 AM	10.0.90.7	%LINK-3-UPDOWN: Interface Port-channel2, changed state to down
15	07.20.2026 11:58:06.560 AM	10.0.90.7	%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to ...
16	07.20.2026 11:58:06.560 AM	10.0.90.7	%LINK-5-CHANGED: Interface Port-channel2, changed state to up
17	07.20.2026 11:58:06.878 AM	10.255.0.13	%HSRP-6-STATECHANGE: FastEthernet1/0 Grp 10 state Standby -> Active
18	07.20.2026 11:58:06.560 AM	10.0.90.7	%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up
19	07.20.2026 11:58:16.306 AM	10.255.0.13	%HSRP-6-STATECHANGE: FastEthernet1/0 Grp 10 state Speak -> Standby

Figure 25. Syslog Service Validation.

11.1.17 NTP Time Synchronization Validation

```
EDGE_PRIMARY#show ntp status
Clock is synchronized, stratum 2, reference is 10.0.50.10
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is EDDE5DE9.00000304 (12:7:5.772 UTC Mon Jul 20 2026)
clock offset is 41.00 msec, root delay is 12.00 msec
root dispersion is 21.63 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll interval is 4, last update was 2 sec ago.
EDGE_PRIMARY#
```

Figure 26. NTP Time Synchronization Validation

11.1.18 Mail Server Operation Validation

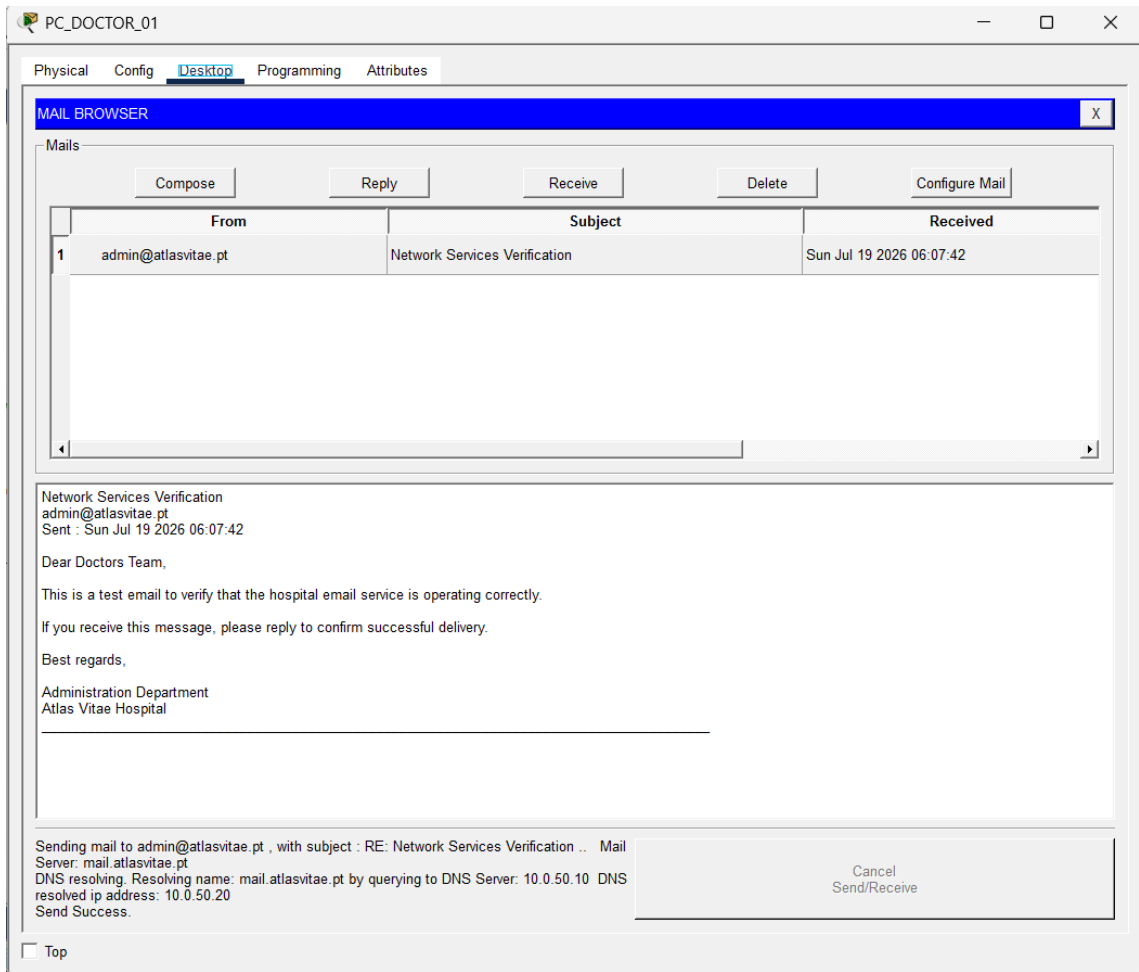


Figure 27. Mail Server Operation Validation 1.

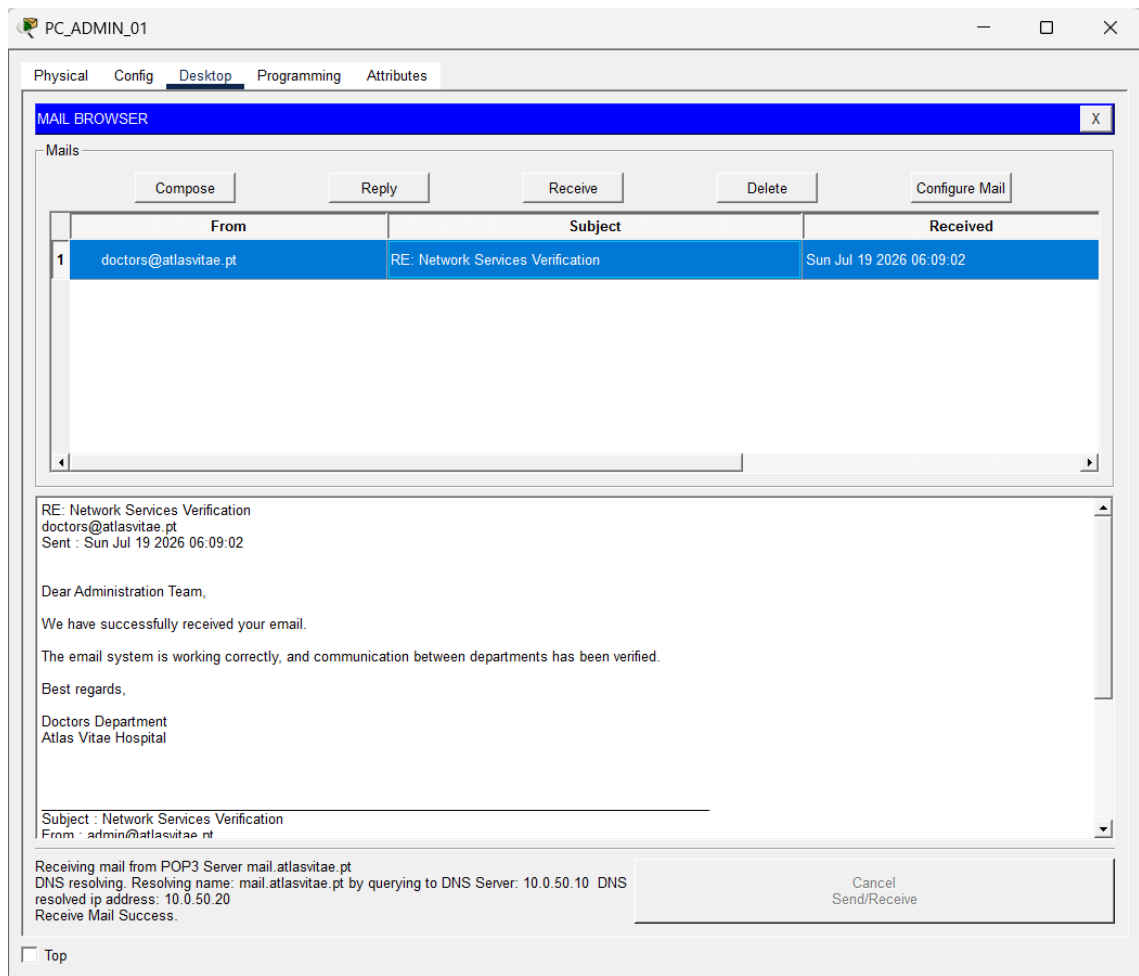


Figure 28. Mail Server Operation Validation 2.

12 Conclusion

The implementation of the **Atlas Vitae University Hospital Enterprise Network** successfully achieved the objectives established during the planning and design phases. A complete Cisco enterprise infrastructure was designed, configured and validated using Cisco Packet Tracer, following modern networking principles and enterprise best practices.

The resulting network provides secure communication between hospital departments through VLAN segmentation, Layer 3 Inter-VLAN Routing and dynamic routing using OSPF. Enterprise services, including DNS, DHCP, FTP, E-mail, Cisco Call Manager Express, internal and public web services, network monitoring, Syslog and NTP, were successfully deployed and integrated into the infrastructure.

Internet connectivity was implemented through a simulated ISP environment using Static NAT, Port Address Translation (PAT) and public IPv4 addressing. Redundancy mechanisms such as HSRP and EtherChannel were incorporated to improve network availability and resilience, while multiple Access Control Lists were configured to enforce communication policies and protect critical resources from unauthorized access.

The entire infrastructure was validated through functional testing, confirming the correct operation of routing, switching, redundancy mechanisms, enterprise services and security policies. The validation results demonstrate that the implemented architecture satisfies the operational requirements defined for the hospital environment.

Beyond its technical implementation, this project provided practical experience in enterprise network design, service integration, routing protocols, security implementation and infrastructure documentation. It represents a realistic enterprise deployment that closely reflects production network environments and demonstrates the application of theoretical networking concepts in a complete, integrated solution.

Overall, the Atlas Vitae University Hospital network constitutes a scalable, resilient and secure enterprise infrastructure capable of supporting modern healthcare operations while providing a solid foundation for future expansion and continuous technological evolution.

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N.º de Aluno: 20240197

Data: 03/06/2026

